

Young, low-income mothers' social relationships and involvement in doula home visiting services

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ARTICLE INFO

Keywords:

Doula
Home visiting
Adolescent mothers
Therapeutic alliance
Service engagement
Latent profile analysis

ABSTRACT

Relationships with family, partners and friends may shape mothers' involvement with home visiting services, and discrepancies exist in the literature regarding whether supportive social relationships are associated with more service involvement or less involvement. Mothers with more support may be more involved in services because their history of support may enable them to build trusting working alliances with service providers (relationships-beget-relationships). Alternatively, mothers with less support may be more involved because they may be in greater needs of services (relationship-compensation). This study examined patterns of mothers' social relationships and associations with the mothers' working alliances (agreement on tasks and goals, bonding) and use of home-visiting services provided by doulas in pregnancy and early postpartum. Data were collected from young, low-income mothers ($N = 147$; mean age = 18.5 years old; 49.7% African American, 41.5% Latina, and 8.8% European American/Other) who participated in doula-enhanced-home-visiting programs and from program administrative systems. Doulas provided home visits starting in pregnancy through six weeks postpartum and provided support during childbirth at the hospital. Eight indicators of mothers' relationships with their current parent figure (usually their mother; PF), the father of their baby (FOB), and general social network were measured. Latent profile analysis identified four relationship profiles: strong support from PF, FOB and others (27%), moderate support overall (40%), moderate support from FOB but alienation from PF (15%), and moderate support from PF but alienation from FOB (18%). Mothers with strong social relationships reported higher agreement with their doula on tasks/goals both during pregnancy and postpartum; mothers with challenging relationships with their parent figure had fewer pregnancy home visits and were less likely to have doula hospital attendance but had similar frequency of postnatal home visits as other classes. Relationship profiles were not associated with bonding with the doula, postnatal service frequency, or early service discontinuation. This study primarily provided support to the relationships-beget-relationships mechanism, but also suggested that mothers with a challenging history with their parent figure may become more involved in services after the birth of their baby. A deep understanding of young mothers' relationship history can help home visitors provide equitable and individualized services to promote family and child wellbeing, especially for mothers with challenging histories with their caregiver.

1. Introduction

Early childhood home visiting refers to a service delivery model in which education, support, and services for pregnant individuals, infants, young children and their families are provided in the family's home by a professional or paraprofessional (Adirim & Supplee, 2013). Evidenced-based home visiting programs can improve health and mental health outcomes for mothers and infants, promote positive parenting, reduce

child maltreatment, enhance family relationships, improve child developmental outcomes and school readiness, and connect families to additional services (Avellar & Supplee, 2013; Peacock et al., 2013). Programs vary in their target population and primary goals. Some provide universal services while others focus on groups with increased vulnerabilities, such as pre-term infants, young mothers, or very commonly, families with lower incomes (Adirim & Supplee, 2013). A large body of literature from developmental psychology, economics and

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<https://doi.org/10.1016/j.childyouth.2023.107228>

Received 24 August 2022; Received in revised form 2 June 2023; Accepted 13 October 2023

Available online 14 October 2023

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neurobiology reveals that children from socioeconomically disadvantaged families lag behind their economically advantaged peers in cognitive and socioemotional development, and that such disparities begin during early life (Yoshikawa et al., 2012). To support and empower disadvantaged families and to foster the healthy development of their young children, home visiting has become a prominent service delivery approach for interventions targeting families with infants and toddlers in under-resourced and underserved communities in the U.S. (Adirim & Supplee, 2013; Duggan et al., 2018).

However, despite a robust body of scientific findings demonstrating the effectiveness of multiple home visiting models in improving child development and family well-being (HomVee, 2022; Kirkland & Mitchell-Herzfeld, 2012; Love et al., 2005; Olds, 2002), the sizes of intervention effects are generally modest. One of the possible reasons is that many mothers do not engage as frequently or deeply with the intervention as the model developers intend (Alonso-Marsden et al., 2013), either in their level of participation or in their emotional connection to the home visitor or the program content (Korfmacher et al., 2008). For example, an evaluation study of 88 home visiting programs from 12 states showed that in the first year of enrollment, about 40% of families received fewer than 50% of the visits expected by their home visiting model (Duggan et al., 2018). Additionally, some mothers have low commitment to programs and poor relationships with their home visitors (Korfmacher et al., 1997). Previous research shows that a variety of factors shape parents' involvement in home visiting services, including the practitioner's and the parent's personal characteristics and previous experiences in health care and social relationships, as well as program and neighborhood characteristics (Bower & West, 2020; McCurdy & Daro, 2001; McNaughton, 2000). This paper aims to explore whether and how mothers' past and current relationships are related to their involvement in doula (a specialized home visitor) services in a randomized controlled trial (RCT) of doula-enhanced home visitation, with implications for broader home visiting services.

1.1. Social relationships and involvement in psychotherapy

Issues of client involvement are relevant to a broad set of social services (Marsh et al., 2012) and have been a focus of study within the mental health field (Norcross & Wampold, 2011). One aspect of service involvement that has gained attention within psychotherapy research is the working alliance between the therapist and the client, which plays an important role in treatment success (Horvath, 2000). Working alliances include three components: bonds (e.g., interpersonal attachments, liking, and trusting), tasks (e.g., agreements between the therapist and the client in what to be done in therapy), and goals (e.g., consensus on outcome expectations between the therapist and the client), and may be influenced by past relationships and unresolved bonds, especially in early phases of therapy (Bordin, 1975).

Specifically, working alliances can be viewed as a special type of attachment bond in which the therapist assumes the role of a consistently responsive attachment figure to foster the clients' trust and provide a secure base for exploration of their feelings, memories and interpersonal environment (Bowlby, 1988; Pistole, 1989). The working models or mental representations of relationships clients developed in other relationships shape their capacity to trust and form attachments with the therapist and to take responsibility in the joint work in therapy (Gelso & Carter, 1985). Clients' childhood experiences with parents have gained special attention due to the continuity of childhood and adult attachment (Raby et al., 2013), and it has been suggested that relationship history with parents may be among the most potent factors that affects clients' ability to form productive working alliances (Malinckrodt, 2000). Empirical evidence has shown that clients with challenging current and past relationships tend to establish weaker working alliances with therapists, while clients with strong relationships and attachment security tend to establish stronger working alliances with therapists, in other words, a positive association between previous

attachment and working alliances with therapists (Diener & Monroe, 2011; Keller et al., 2010; Mikulincer et al., 2013). However, it is worth noting that working alliances are also shaped by factors beyond past attachment, such as therapists' techniques (collaboration, emphasizing confidentiality vs. pushing the client to talk) (Creed & Kendall, 2005) and clients' trauma symptoms (Bovard-Johns et al., 2015). Research also suggests that clients with past trauma and insecure attachment can be involved in building trusting working alliances when their feelings and experiences are normalized and validated, and when they gain new insights in understanding their past and its emotional impact and are empowered to manage current lives (Knight, 2015).

1.2. Social relationships and involvement in home visiting services

Evidence suggests that a home visitor may act as an important attachment figure for mothers enrolled in services, providing comfort and support during pregnancy and the postpartum years (Bilukha et al., 2005). Many mothers enrolled in home visitation have experienced challenges or discordance in family relationships during childhood and with current caregivers or partners (Bosquet & Egeland, 2001). Mothers' relationships with their family or partners can shape their emotional engagement and use of home visiting services. Studies have shown that mothers from supportive families are able to easily develop bonds with their home visitors and engage with services (Beasley et al., 2018), while mothers with low family support can develop either strong or weak bonds (Brookes et al., 2006). One study found that mothers who felt rejected by their parents and had suffered abuse from past partners reported that they were unwilling to address problems with home visitors as they learned to be independent from hurtful relationships (Brookes et al., 2006). Another study reported similar findings: Some mothers who grew up experiencing abusive relationships perceived health professionals to be authoritative and bossy, like their parents, and tried to avoid negative judgments by not using services (Hanna, 2001). Some studies measured mothers' social relationships through the lens of adult attachment, finding a positive association between attachment security and trust with service providers and service participation (Korfmacher et al., 1997; McFarlane et al., 2010).

Although the dominant view in the home visiting literature is that strong relationships with family and partners (or secure attachment) can lead to increased levels of service involvement, an alternative view is that mothers with less support or more difficult relationships may have fewer readily available helping resources (e.g., advice, information, and instrumental aid) and experience more stress. Therefore, they may have higher needs for support provided by home visitors (Birkel & Reppucci, 1983), facilitating the growth of working alliances and increasing their use of services. A few methodologically strong quantitative studies have shown that mothers who faced more psychosocial risks, such as low levels of social support, were more involved in services. Researchers argued that this association was due to home visiting programs' strong emphasis of a close and trusting relationships, which may be especially important for isolated mothers with limited social support, or to the possibility that service providers were more persistent with these mothers whom they perceived as having greater needs of services (Ammerman et al., 2006; Duggan et al., 2000; Miller et al., 2019). Additionally, one qualitative study suggested that differences in program involvement may also depend on the stage in the services (e.g., initial vs. later), in addition to mothers' social relationships with family and partners: Those with successful social relationships tended to develop trust with home visitors quickly, and those with rejecting and abusive relationships may develop trust after working with home visitors for extended periods (Brookes et al., 2006).

An insight provided by Olds and Korfmacher (1998) may be useful to understand the discrepant findings regarding social relationships and home visiting involvement. Their findings suggested two different processes that contribute to home visiting involvement: (1) Mothers with very high psychological resources are able to receive more visits, even

though their need for services is lower; (2) Mothers with low psychological resources likely experience more times of crisis and have greater need for services, especially “a person to talk to who really cared” (Pharis & Levin, 1991), which allows them to be more open to home visiting support.

Although psychological resources and social relationships are not equivalent, these two mechanisms could be underpinning the association between social relationships and service involvement. One can be labeled the *relationships-beget-relationships* mechanism: Mothers with strong supportive social relationships are more involved in services since these relationships serve as models for developing new, trusting connections with providers more quickly, and existing support systems enable mothers to create time and space for engaging with services and being present during visits. The second is the *relationship-compensation* mechanism: Mothers with less support and more challenging social relationships may engage in services more since they are in greater need of psychosocial support. Their involvement in in home visitation may compensate for the lack of strong social support in their life, especially when home visitors are actively targeting them for service delivery and building strong working alliances.

Although the *relationships-beget-relationships* and *relationship-compensation* mechanisms make seemingly opposite predictions regarding the association between social relationships and service involvement, they could both hold true, depending on the parameters of program involvement one is considering. In particular, it is important to examine not only the amount of services received, but also the quality of the relationship between mother and home visitor. It is also important to examine both initial engagement with programs and how program involvement evolves over time. This study aims to explore if the two mechanisms may be applicable for different aspects of program involvement (quantity and quality) and at different times in program involvement (initial and later). Given that the overall body of literature is relatively small and that no definitive theories have prevailed, we do not have specific hypotheses regarding which mechanism holds true at which timepoint. Rather, our study is exploratory in nature, guided by the insights from Olds and Korfmacher (1998) and Brookes et al. (2006).

1.3. The social relationships of young mothers: A multidimensional construct

Because many home visiting models serve first-time low-income mothers (e.g., Healthy Families America and Nurse-Family Partnership), young mothers, including teenage mothers, often make up a significant proportion of home visiting participants. The family support landscape of young mothers can be very different from that of older mothers, and the role of grandmothers or other parent figures may be particularly important.

Young mothers often live with their own mother or another adult parent figure (PF), with co-residence rates over 70% for adolescent participants in some home visiting programs (Hans et al., 2018; Mistry et al., 2016), and they often identify parent figures as their primary source of social support (DeVito, 2007). In addition to receiving tangible financial and childcare support from parent figures, many young mothers also report being emotionally close to their families, particularly to their mothers, during their transition to motherhood (Hunt et al., 2011), although conflict and tension between young mothers and their mothers are common as they juggle the demands of parenting an infant (East & Felice, 2014). In contrast, some young mothers receive little support from their families or support that is adversarial, particularly those who grew up experiencing abusive relationships with their family (Aparicio et al., 2015; Hanna, 2001; SmithBattle & Leonard, 2014). Young mothers' relationships with their family, particularly with their own mother, have been shown to shape their confidence as parents and their quality of their interactions with their infants (Chase-Lansdale et al., 1994; Oberlander et al., 2007; Seay et al., 2016; Voight et al., 1996). Importantly for the issue of mothers' involvement in home

visiting programs, family members may be important gatekeepers determining whether home visitors are allowed in the household and are important decision makers regarding childrearing practices (Daro & Harding, 1999; Dunifon, 2013).

The relationships between teenage mothers and their male partners may be particularly fragile. American teenage mothers typically are unmarried and usually do not live with the infant's father (Carlson & McLanahan, 2010). Young fathers, like teenage mothers, often start parenthood with backgrounds of economic insecurity and limited educational achievement (Bamishigbin et al., 2019) that present numerous barriers to fulfilling the traditional partner and father roles as an economic provider (Dukes & Palm, 2019; Florsheim & Moore, 2020). However, despite these challenges, many young fathers are highly involved with the mothers and young children (Edwards et al., 2012; Howard et al., 2006; Kalil et al., 2005), particularly in the time frame around the birth (Bellamy et al., 2015; Easterbrooks et al., 2016; Gee & Rhodes, 2003), although father involvement declines notably over the first year after birth (Thullen et al., 2012). Although often less involved that older men in the day to day lives of their partners and infants, support from young fathers is important in shaping teenage mothers' emotional well-being during pregnancy and postpartum.

Existing quantitative studies examining the association between social relationships and home visiting involvement have typically considered social relationships as a unidimensional construct that ranges on a continuum from strong to poor (e.g., Wen et al., 2016). Yet mothers' close relationships differ with respect to many important dimensions, including attachment quality of relationship, type of support provided (e.g., emotional versus tangible), timing of relationship in life course (e.g., childhood versus current relationships), and the degree to which relationships are sources of conflict, stress, and/or disagreements (Pierce et al., 1996). Moreover, mothers' support systems are networks of multiple people, who may vary in the type and quality of support they provide the mother.

Qualitative studies suggest that adolescent mothers vary in the pattern of supportive relationships they experience. For some mothers, there may be parallels across relationships. For example, some young mothers have strong support from their families, their infants' fathers and his family (e.g., Mollborn & Jacobs, 2015), other young mothers were raised with experiences of family violence and continue to experience violence from their male partners (Bekaert & SmithBattle, 2016; Ethier, 2022), and a few young mothers are largely lacking in support from their personal networks (e.g., Schrag & Schmidt-Tieszen, 2014). Alternatively, young mothers may have divergent quality of relationships with family and partners. For example, some young mothers parent their children with considerable help from their family but the absence of their infants' fathers (e.g., Edin & Kefalas, 2011), and for other young mothers the pregnancy leads to ruptures with their own family but strong commitment from the father of their infant (e.g., Aparicio et al., 2015). These are examples of typologies of relationships whose importance may not be captured in quantitative studies if relationship features are measured as isolated variables. A broader literature on adolescents and young adults suggests that it is fruitful to consider teenagers relationships, not in terms of single dimensions, but in terms of typologies or profiles that capture patterns of support (Garthe et al., 2020; Levitt et al., 2005).

This study conceptualizes supportive relationships as a multidimensional construct encompassing four different sources of support (childhood caregiver, current parent figure (PF), father of the baby (FOB), general social network), two types of support from PF and FOB (emotional and tangible), and the extent to which relationships are strained with childhood caregiver, PF and FOB. We will use latent profile analysis, a person-oriented analytic method (Sterba & Bauer, 2010), to identify profiles or subgroups of social relationships and to examine their association with measures of program involvement. The goal of these analyses will be to seek evidence for the *relationships-beget-relationships* and/or *relationship-compensation* mechanisms. Findings can

help home visiting service providers gain a deep and holistic understanding of the multidimensionality of young mothers' social relationships and provide insights in tailoring service design and implementation to meet the needs of participants from different relationship backgrounds.

1.4. Doula-enhanced home visiting services

This study leveraged data from an RCT of doula-enhanced home visiting programs. Each mother in the intervention group of the RCT worked with a team of two service providers—a community doula whose work focused on providing specialized home-based support and education during the pregnancy (starting around 6 months of pregnancy on average) and early postpartum period, and a home visitor (family support worker or parent educator) whose work focused on providing support for parenting and child development and general family support throughout infancy and early childhood. All doulas and home visitors had deep roots in their communities, most were from the same ethnic/racial background as their clients, and some had been adolescent mothers themselves. Both doulas and home visitors completed the foundational training required by their program models, and doulas also completed additional training provided through the Ounce of Prevention Fund. Developing strong relationships and trust with families was a core element in home visiting models (Bernstein, 2002).

In this program model, during pregnancy and early postpartum, mothers were visited weekly by a doula, home visitor, or both together, with the doula being the primary service provider. Doulas' home visits included preparing mothers for labor and delivery and discussing lactation, early bonding with the infant, newborn care, and mothers' health and mental health. Doulas came to the hospital during labor and delivery to offer the mothers physical comfort, emotional support, and breastfeeding counseling. By focusing on and attending childbirth, doulas are engaged in an intimate form of contact with mothers (Humphries & Korfmacher, 2012; Korfmacher & Humphries, 2015). Doulas continued to make regular weekly home visits after the birth through six weeks postpartum, during which time they focused on mothers' postpartum health and baby's early development. Home visitors increased the frequency of their visits after the birth and became the primary service provider after six weeks postpartum, offering services throughout infancy and toddlerhood. This paper focuses on mothers' service involvement with doulas, since doulas were the primary service provider in this home visiting model during the critical early months of program engagement.

1.5. Study questions

This study aims to examine two research questions. First, what types of social relationships (i.e., relationship profiles) do young, low-income mothers experience with their support network? Second, how are different relationship profiles associated with mothers' involvement (engagement and participation) in doula-enhanced home visiting services in pregnancy and early postpartum, and how does involvement change from pregnancy to early postpartum?

2. Method

2.1. Sample and procedure

Data were from a randomized controlled trial (RCT) of doula home visiting services provided at four program sites in high-poverty Illinois communities ($N = 312$). Two program sites were in a large city, and two in smaller urban areas. Programs were not demonstration programs for only research purposes but were from a network of state-funded home-visiting programs. For a detailed description of the RCT study, please refer to Hans et al. (2018). Pregnant women were randomized to receive either case management services or doula-enhanced home visiting

services from programs that incorporated doulas into existing evidence-based models of [Healthy Families America \(n.d.\)](#) or [Parents as Teachers \(n.d.\)](#). Since the main interest of this paper is on mothers' use of home visiting services and relationships with doulas, only data from mothers in the intervention group were used ($n = 156$). The final analytic sample size was 147 ($M = 18.5$ years, $SD = 1.88$). [Table S1](#) in the [Supplementary Materials](#) presents the sample characteristics. Nine mothers were excluded from all analyses since they had missing data on indicators of mothers' social relationships (reasons can be found in [Table S2](#)), and only those with complete data of indicators (details of indicators are in [Section 2.2.1](#)) were kept.

Demographic, psychosocial and working alliance data were gathered from interviews with the mothers at study enrollment during pregnancy (i.e., baseline), 37 weeks of pregnancy, and 3 weeks postpartum. Interviews were usually conducted at the mother's home and administered in the mother's preferred language—English or Spanish. At the baseline interview, mothers were asked to choose from five racial/ethnic categories (i.e., Black/African American, Latina/Hispanic, White/European American, bi-/multi-racial, and "other"). Mothers who answered "bi-/multi-racial" were then assigned to one of the other four categories for analyses based on the specific ethnicity they mentioned and/or information they provided about their family and community. In order to identify the mother's parent figure, she was asked the question at the baseline interview, "Is there someone you would say is like a parent to you now, who watches out for you and is responsible for your well-being?" About 80% of the primary childhood caregivers and the current parent figures were the biological mother, 10% were grandmothers, and 5% were biological fathers. The childhood caregiver and the current parent figure were the same for most of the participants (91.8%, $n = 135$). Maternal depressive symptoms were measured using the Center for Epidemiological Studies-Depression scale (CES-D; [Radloff, 1977](#)), a self-report questionnaire that assesses symptoms experienced in the past week and provides a cut-off score (≥ 16) indicating risk for clinical depression (Cronbach's alphas were 0.82, 0.85, and 0.85 at the three waves). Administrative data of service records were available from management information systems (MIS) used by the home visiting programs regarding date of service, time service began, length of service (in minutes), person delivering service, and location of service (home, hospital, phone). Data from the MIS systems were crosschecked against mother report for accuracy.

2.2. Measures

2.2.1. Mothers' social relationships

The means of the following ten indicators were used to identify the profiles of social relationships reported at the time when mothers enrolled in the study: perceived rejection from childhood caregiver; trust, communication, alienation from, and received support from current parent figure; trust, communication, alienation from, and received support from father of the baby; and quality of general social relationships.

Perceived rejection from childhood caregiver. Three subscales from the Parental Acceptance-Rejection Questionnaire (PARQ; [Rohner, 2005](#)) were used to measure the extent to which mothers perceived acceptance and rejection from her primary caregiver in childhood: warmth and affection, hostility and aggression, and indifference and neglect. Mothers responded to 16 items on a 4-point Likert scale (4 = "almost always true" 1 = "almost never true"). Sample statements were "when I was between 6 and 12, my [childhood caregiver] ... totally ignored me," "took an active interest in me," and "frightened or threatened me when I did something wrong." The PARQ has been well validated internationally ([Rohner & Ali, 2016](#)). The Cronbach's alpha of the 16 items was 0.85. Higher scores indicate higher levels of perceived rejection in childhood.

Trust, communication, and alienation from current parent figure and the father of the baby. The Inventory of Parent and Peer Attachment (IPPA;

Armsden & Greenberg, 1987) was used to measure participants' perceived attachment with their current parent figure and the father of the baby. Mothers responded on a 5-point Likert scale (5 = "almost always true" 1 = "almost never true") across three domains—trust, communication and alienation, each of which had three items. Sample statements included "my [parent figure/baby's father] respect my feelings," "if he/she knows something is bothering me, he/she asks me about it," and "he/she doesn't understand what I'm going through these days." Cronbach's alpha for the trust, communication, and alienation subscales were 0.80, 0.78, and 0.63 for parent figure, and 0.90, 0.90, and 0.69 for the father of the baby. Higher scores on each subscale indicated more trust, better communication, and greater levels of alienation.

Received support from current parent figure and the father of the baby. The mean score of a six-item scale developed and used with a similar group of mothers (Edwards et al., 2012; Thullen et al., 2014) was used to measure support provided by the parent figure and the father of the baby. Participants responded on a 5-point Likert scale (5 = "very often" 1 = "never"). The six items were: during the past 3 months how often has your [current parent figure/baby's father] "bought you things you needed or given you money to buy things you need?" "listened to your worries, concerns, or problems?" "answered your questions and given you advice when you wanted it?" "done something with you that you both enjoy?" "given you help you needed, like with transportation, schoolwork, a household chore, or something like that?" and "helped you make plans for after your baby is born?" Cronbach's alpha for the six items was 0.88 for the parent figure, and 0.94 for the father of the baby. Higher scores indicated more received support.

Quality of general social relationships. The quality of participants' general social relationships was measured by the Social Provisions Scale (Cutrona & Russell, 1987), which included six subscales—attachment, social integration, reassurance of worth, reliable alliance, guidance, and opportunities for nurturance. Participants responded on a 4-point Likert scale (4 = "strongly agree" 1 = "strongly disagree") to each item. Sample statements included "there are people I can depend on to help me if I really need it," "there is someone I could talk to about important decisions in my life," and "I have close relationships that provide me with a sense of emotional security and well-being." Cronbach's alpha for the 24 items was 0.84. Higher scores indicated higher quality of general social relationships.

2.2.2. Service involvement

Eight outcomes of service involvement were of interest: four were related to engagement, including bonding and agreement about tasks/goals at 37-weeks pregnancy and 3-weeks postpartum, and four were related to participation, including the frequency of prenatal and postnatal doula visits, doula hospital attendance, and early discontinuation of services.

Bonding and agreement about tasks/goals with the doula. To measure the quality of the relationship with the doula, a modified version of Tracey & Kokotovic's (1989) 12-item Working Alliance Inventory WAI short form (WAI-S) was used at both the 37-week prenatal interview and the 3-week postnatal interview. Mothers responded to 12 items on a 4-point Likert scale (4 = "Always" 1 = "Never"). WAI-S is composed of three four-item subscales—bonding, tasks, and goals. Sample items include "I appreciate all my doula has done for me" and "My doula and I trust each other" from the bonding subscale, "my doula and I agree about the steps to be taken to make things better for me and my baby" from the tasks subscale, and "my doula and I are working toward goals we both agree on" from the goals subscale.

Studies have shown that a two-factor structure better captures the working relationship since clients usually have trouble distinguishing between tasks and goals (Gessnitzer & Kauffeld, 2015). Therefore, the average score of eight items from the tasks and the goals subscales were used to measure participants' agreement about tasks/goals in their working relationship with the doula. The Cronbach's alpha of the eight

task/goal items was 0.76 at 37-weeks pregnancy and 0.83 at 3-weeks postpartum. The average score of the four items from the bonding subscale was used to measure participants' emotional bond and closeness with the doula, and Cronbach's alpha of the four items was 0.88 at 37-weeks pregnancy and 0.89 at 3-weeks postpartum. Since bonding and agreement about tasks/goals were highly skewed to the left, they were first reflected and then transformed using logarithm to alleviate skewness in the following analyses (Chatterjee & Hadi, 2015).

Frequency of prenatal and postnatal doula visits. In the administrative system, the date of each doula service was recorded by the doulas. First, the numbers of total prenatal and postnatal doula visits were calculated using the administrative data. The administrative data were cross-checked against participants' self-report data. Discrepancies between the two sources were minor and the administrative data were used for analyses. Because mothers were enrolled in the program for different amounts of time prenatally, home visits were converted to visits per week of enrollment in the program. For prenatal visits, the number of pregnancy visits were divided by the number of weeks from the first doula home visit to the baby's date of birth. Since all participants were eligible for postpartum doula services until six weeks postpartum, the number of postnatal visits was divided by six.

Doula hospital attendance at birth or shortly after birth. At the 3-week postpartum interview, participants responded to the questions "was your doula present when you gave birth?" "If no, was another doula present for the birth?" and "Did your doula visit you in the hospital the next day?" Mothers who responded "yes" to at least one of the three questions were coded 1 on the doula hospital attendance outcome. Several mothers gave birth very quickly before their doula arrived at the hospital, and some mothers who had caesarean sections had doulas present at the hospital but were not allowed to have the doula present in the operating room. For these reasons, doulas who were at the hospital providing support, even if not present for the actual birth, were counted as "yes" on the hospital attendance outcome. The self-report data were cross-checked against the service record in the administrative system in which "doula attended birth" or "no doula attended birth" was noted on the baby's date of birth, or "doula hospital service" was noted shortly after the birth. No noticeable differences were found. There were 13 participants who did not have interview data at 3-weeks postpartum, and their doula hospital attendance was filled solely based on information from the administrative system.

Early discontinuation of services. In the administrative system, if a participant's last doula service date was before 6-weeks postpartum, they were coded as 1 in the outcome of early discontinuation of services. Otherwise, they were coded as 0.

2.3. Analysis methods

Analyses for this study followed three steps. First, latent profile analysis (LPA) was conducted to generate profiles of social relationships in the analytic sample using ten continuous indicators measured at the baseline interview. LPA and latent class analysis (LCA) are similar since both assume a categorical latent variable (i.e., the latent class). However, continuous indicators are used in LPA, while categorical indicators in LCA (Dziak et al., 2016). Mothers who shared similar characteristics in their social relationships were grouped into one latent class. Models with two to four classes were examined. Both substantive interpretability and statistical criteria were taken into account, including Bayesian Information Criterion (BIC; Schwarz, 1978) (model with smaller BIC was preferred), entropy (a value closer to 1 was preferred), and Vuong-Lo-Mendell-Rubin likelihood ratio test (significant *p*-value indicated the model with *k* classes fits better than *k*-1 classes). The class prevalence, class-specific means and standard deviations of each indicator, and the probability of belonging to each latent class for each mother were estimated. Mothers were then assigned to the latent class to which they had the highest probability of belonging (i.e., maximum probability assignment). Since the assigned class membership may

contain classification error, a weight equal to the value of the maximum probability was created and used in the following analyses (i.e., probability-weighted regression; Cheng, 2012; Kamata et al., 2018; Nese et al., 2017).

Second, weighted regressions were used to examine whether the mothers' supportive relationship profiles were associated with the eight outcomes of service involvement. Weighted ordinary least squares regressions were used for continuous outcomes and logistic regressions were used for dichotomous outcomes. Other maternal characteristics were not controlled for since they are already captured in the latent profile membership.

Third, the longitudinal changes in the working alliance from the 37-weeks pregnancy to 3-weeks postpartum and the change in the service frequency from the prenatal to postnatal period were examined via paired *t*-tests within each class. The first step was conducted in MPlus 8.0 (Muthén & Muthén, 2017), and the other two steps were conducted in Stata 15.0 (StataCorp, 2017).

3. Results

3.1. Identifying profiles of social relationships using latent profile analysis

After considering the statistical fit indices and the substantive interpretations, a four-class latent profile analysis solution was chosen (details of model fit indices and how the four-class was chosen can be found in Table S3 in Supplementary Materials). Table 1 shows the means of each indicator across the four classes. Mothers classified in Class 1, comprising 27% of the sample, experienced the lowest levels of rejection in childhood, the least alienation from their current parent figure (PF) and father of the baby (FOB), the highest levels of trust, communication with, and support from their parent figure and the father of the baby, and the highest quality of general social relationships. This class is labeled as *strong support overall*. Class 2 is the largest subclass in the sample (40%), and mothers classified in this class scored moderately on all ten indicators. Therefore, Class 2 is labeled as *moderate support overall*. Mothers in Class 3 comprised the smallest class in the sample

Table 1
Estimated Means of the Ten Indicators of Relationship Profiles (N = 147).

	Class 1Strong support (n = 39, 27%)	Class 2Moderate support (n = 61, 42%)	Class 3Alienation from PF (n = 21, 14%)	Class 4Alienation from FOB (n = 26, 18%)
Perceived rejection from childhood giver (1–4)	1.10	1.38	1.63	1.42
Parent figure (1–5)				
Trust	4.85	4.26	2.82	4.51
Communication	4.73	4.03	2.58	4.33
Alienation	1.63	2.55	3.15	2.67
Received support	4.61	4.22	2.88	4.51
Father of the baby (1–5)				
Trust	4.97	4.08	4.40	2.02
Communication	4.74	3.79	4.42	1.52
Alienation	1.59	2.59	2.30	3.35
Received support	4.73	3.90	4.35	1.39
Quality of general social relationships (1–4)	3.51	3.20	3.13	3.23

Note. 1–4 refers to a 4-point Likert scale, and 1–5 refers to a 5-point Likert scale. Highest value of each indicator is in bold and lowest value is in italics. PF = parent figure. FOB = father of baby. Class 3 = Moderate support from FOB but alienation from PF. Class 4 = Moderate support from PF but alienation from FOB.

(15%) and reported the highest levels of rejection in childhood and current alienation from their parent figure, and the lowest levels of trust, communication with and support from their parent figure. Despite a difficult relationship with their parent figure, they had a relatively strong relationship with the FOB (e.g., they reported scores around 4.5 on the 5-point scales of trust, communication with and support from FOB). Therefore, Class 3 is labeled as *moderate support from FOB but alienation from PF* (shortened as *alienation from PF*). Mothers in Class 4 (18%) demonstrated the opposite relationship profile as that of Class 3. Class 4 reported the highest level of alienation from FOB and the lowest levels of trust, communication with, and support from FOB. Despite a difficult relationship with the FOB, they had moderated levels on the rejection from their childhood caregiver and the four indicators related the current PF (i.e., trust, communication, alienation and received support). Therefore, Class 4 is labeled as *moderate support from PF but alienation from FOB* (shortened as *alienation from FOB*). The quality of general social relationships was higher in the *strong support overall* class (3.51) than the other three classes, which had relatively similar levels (3.20, 3.13, and 3.23). (Differences in maternal characteristics by latent profile membership can be found in Supplementary Materials Table S4.).

3.2. Differential service involvement with doula across the four latent profiles

Table 2 presents the service involvement for each of the four classes. Overall, mothers in the four classes reported strong working alliances with their doula (around 3.5 on a 4-point scale). The majority of mothers had a doula present during childbirth (ranging from 74% to 93% across the four classes), received about one doula visit per week both during pregnancy and after the birth, close to the program target, and did not discontinue services before six weeks postpartum (attrition rates ranging from 11% to 20% across the four classes).

Latent profile membership was not associated with bonding with the doula at 37-weeks of pregnancy or 3-weeks postpartum, postnatal service frequency, or service discontinuation before 6-weeks postpartum. However, it was significantly associated with agreement on tasks/goals with the doula at both 37-weeks of pregnancy and 3-weeks postpartum, doula hospital attendance, and prenatal service frequency.

Agreement on tasks/goals with the doula (engagement). Although all four classes reported a decrease in agreement on tasks/goals from 37-weeks of pregnancy to 3-weeks postpartum (Fig. 1), paired *t*-tests showed that only the decrease for the *alienation from PF* and *strong support* were significant ($p < 0.05$). Mothers in the *strong support* reported higher agreement on tasks/goals with the doula than mothers in the *moderate support* and *alienation from FOB* at 37-weeks of pregnancy, and mothers in the *alienation from PF* at 3-weeks postpartum.

Doula hospital attendance (participation). Compared to mothers in the *alienation from PF*, those in the *moderate support* were more likely to have a doula attending their birth or visiting them in the hospital shortly after birth (OR = 5.93, 95% CI [1.39, 25.35]).

Service frequency (participation). Mothers in the *strong support* and the *alienation from FOB* had higher service frequency than mothers in the *alienation from PF* class during pregnancy. However, all four classes had similar levels of service frequency postpartum. Fig. 2 presents the changes in service frequency for each class, and paired *t*-tests show that mothers in the *alienation from PF* class experienced a significant increase in their service frequency from the prenatal to postnatal period while mothers in the other three classes remained stable in their service frequency over time.

4. Discussion

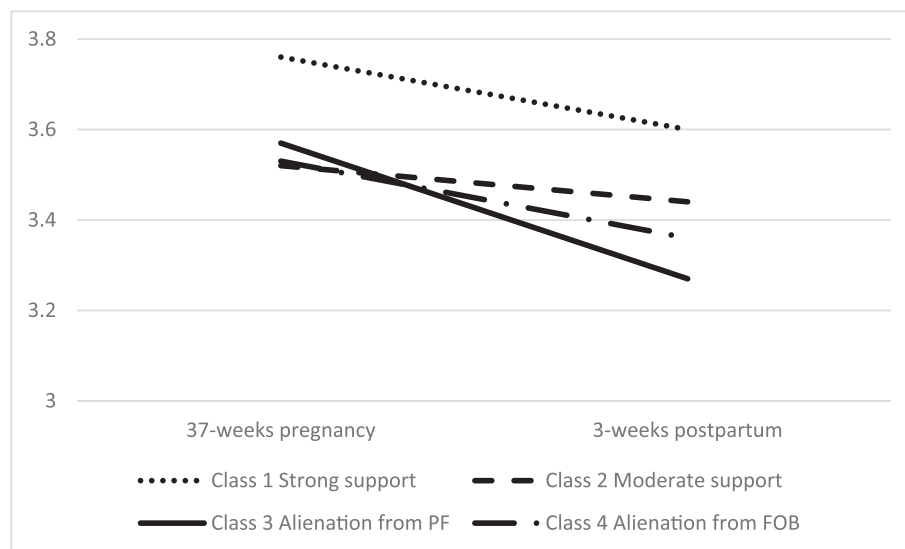
This study identified four qualitatively different patterns of social relationships of young, low-income mothers enrolled in doula-enhanced home visiting services: *Strong support*, *moderate support*, *moderate support from the father of the baby but alienation from parent figure*, and *moderate*

Table 2

Service Involvement by Latent Profile Membership.

	Class 1Strong support (n = 39, 27%)	Class 2Moderate support (n = 61, 42%)	Class 3Alienation from PF (n = 21, 14%)	Class 4Alienation from FOB (n = 26, 18%)	Significantcomparisons
Working alliance with doula					
Bonding					
37-weeks pregnancy (n = 117)	3.75	3.73	3.63	3.67	
3-weeks postpartum (n = 124)	3.75	3.68	3.56	3.56	
Changes in bonding	n.s.	n.s.	n.s.	n.s.	
Agreement on tasks/goals					
37-weeks pregnancy (n = 117)	3.76	3.52	3.57	3.53	1 > 2 **
3-weeks postpartum (n = 124)	3.60	3.44	3.27	3.36	1 > 4 *
Changes in agreement on tasks/goals	p < 0.05	n.s.	p < 0.05	n.s.	1 > 3 **
Doula hospital attendance (n = 138)	33(86.8%)	54(93.1%)	14(73.7%)	21(91.3%)	2 > 3 *
Service frequency (doula visits per week)					
Prenatal period (n = 129)	0.91	0.86	0.61	0.89	1 > 3 **
Postnatal period (n = 135)	0.81	0.85	0.97	0.89	2 > 3 †
Changes in service frequency	n.s.	n.s.	p < 0.01	n.s.	4 > 3 *
Early discontinuation of services (n = 135)	7(20.0%)	8(13.8%)	2(11.1%)	4(16.7%)	

Note. Means of bonding and agreement on tasks/goals for each class were presented. For doula hospital attendance and early discontinuation of services, the number of participants with a value of 1 on these two outcomes in each class was presented (percentages in parentheses). † p < 0.1; * p < 0.05; ** p < 0.01; n.s. not significant.

**Fig. 1.** Changes in Agreement on Tasks/Goals with the Doula from Pregnancy to Postpartum.

support from parent figure but alienation from the father of the baby. The variability and patterns revealed via latent profile analysis supported the different types of support (e.g., with and without strong support from families) found in prior qualitative studies (Beasley et al., 2018; Brookes et al., 2006). Overall, our findings suggest strong engagement and participation with respect to the program. Mothers across the four classes received about one doula visit per week during pregnancy and early postpartum, which was close to the program target; the majority of mothers had a doula present at the birth; and few discontinued services before six weeks postpartum. Mother ratings of the working alliance with their doula were also quite high, which is consistent with findings from other home visiting studies (Korfmaier et al., 2007; Wen et al., 2010). Bonding with doulas across the four classes did not differ significantly and was high overall, which may be partly due to the limited power to detect statistical significance and partly because

mothers liked doulas and appreciated working with them overall. However, there were differences in mothers' involvement with services depending on their social relationship backgrounds and the time points. In particular, mothers who reported strong support across their social relationships developed stronger working alliances with their doulas. Mothers in difficult relationships with their own parent showed lower service participation rates during pregnancy and childbirth, though not postpartum.

Overall, our findings provided evidence to the relationship-begets-relationship mechanism. Mothers in the *strong support* class reported feelings of acceptance from their caregiver during childhood and high levels of trust and communication in current relationships. Like mothers in the other three classes, they liked and felt close to their doulas. But they also perceived their desired goals and approaches to achieve these goals to be more aligned with their doulas' goals than did mothers in the

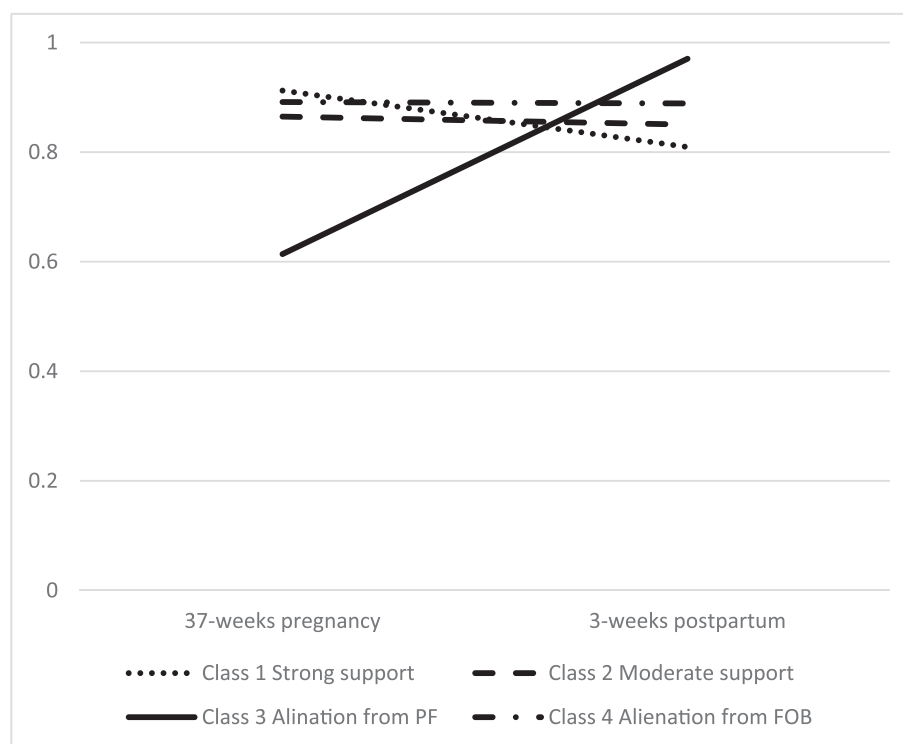


Fig. 2. Changes in Service Frequency (number of doula visits per week) by Latent Profile Membership.

other three classes. These mothers' experiences of supportive social relationships may make it easy to build trust with their doulas, to express their views openly to their doulas, and to negotiate and work out differences with their doulas, which is consistent with the relationship-begets-relationship mechanism. Mothers in the *alienation from PF* class reported the highest level of rejection from caregiver in childhood and the lowest level of support from their current parent figure. They received less intense home visits prenatally and were less likely to have their doula attend the birth at the hospital compared to mothers in the other classes. It is possible that unlike mothers in the *strong support* class, mothers in the *alienation from PF* class faced challenges in building trust and working with doulas since their relationship histories may prime them to distrust new relationships (Brookes et al., 2006; Levine, 2013). They may worry that the doula, like their own mother, might not provide them with the needed comfort and understanding, especially at a time when they feel particularly vulnerable, such as the labor and delivery. However, it is worth noting that future research should explore other possible factors and processes that may explain the association between the *alienation from PF* profile and service involvement. For example, home visits for young mothers often occur in their parent figures' homes. Young mothers who have difficult relationships with their parent figures, may lack a stable location for home visits or may find it difficult to get permission from the parent figure for a doula to visit. It may also be that for teenage mothers, an engaged parent figure is important for reminding young mothers of upcoming home visits or to contact doulas at the onset of labor.

The longitudinal changes in working alliances and service frequency from 37 weeks of pregnancy to 3-weeks postpartum revealed interesting patterns. All four classes of mothers experienced a decrease in agreement on tasks and goals, with it being statistically significant in the *alienation from PF* class and the *strong support* class. This decreasing trend is somewhat consistent with the findings from a different study of community-based doulas: Mothers' engagement (e.g., interest and initiative in problem solving with doulas, and mastery of the materials and ideas discussed in doula visits) showed a slight decrease after the baby was born (Wen et al., 2016). This decrease may be due to several

possible reasons. First, mothers got to know doulas better over the course of services and became more aware of their agreements and disagreements. As they gradually strengthened their relationships with doulas, they may also have felt more empowered and comfortable to voice disagreements. Second, it may be easier to agree on issues related to pregnancy health than on postpartum issues, such as newborn care, breastfeeding, infant sleep, and other aspects of early parenting since the family often plays an important role in the latter while the former may be deemed as in the realm of health professionals (Hwang et al., 2016; Martin et al., 2013; Moon et al., 2019; Sanders & Crozier, 2018; Sink, 2009). Future research can further examine whether the decrease in agreement on tasks and goals can be replicated and if so, whether the hypothesized reasons may hold. However, despite the decrease, it is worth noting that the average score of agreement on tasks and goals for all four classes still indicated a relatively good relationship at 3-weeks postpartum (above 3.3 on the 4-point Likert scale).

The longitudinal findings about mothers in the *alienation from PF* class are particularly notable because they provided some emerging evidence to the relationship-compensation mechanism. Consistent with the other classes, mothers in this group showed a decrease in agreement on tasks and goals with their doulas over time, but they also paradoxically showed an increase in doula visits after the birth (weekly, on average) compared to pregnancy (about one visit every other week). They received fewer prenatal visits per week compared to the other groups but received postnatal visits at a similar frequency. Additionally, they were not at any higher risk of discontinuing services before six weeks postpartum. It is likely that mothers in the *alienation from PF* class may be somewhat hesitant to trust doulas during pregnancy given their history of difficult relationships, but the relationship-based interventions may enable them to gradually let down their guard, reevaluate and modify their initial distrust of doulas to some extent, develop a closer relationship, and increase their service frequency postpartum - a process that may take them a longer time than for mothers in other classes (Brookes et al., 2006). Additionally, an increased need for support to meet the challenges of intensive newborn care and motherhood postpartum may also play a role: Mothers may

deliberately choose to rely on support from doulas to compensate for the lack of support from their families, especially their own mothers - usually a primary source of support (Clemmens, 2003; DeVito, 2007), and doulas may actively target these mothers whom they perceive are at risk to help them cope with the challenges arising from motherhood. Future research involving the perspectives of mothers and doulas and investigating the influence of the newborn baby may further elucidate reasons for longitudinal changes in service involvement among mothers in the *alienation from PF* class.

5. Limitations and future research

Interpretation of the findings needs to take into consideration limitations of this study. First, the four profiles of social relationships were generated using a sample of young, urban, low-income mothers participating in doula home visiting services. Generalizations to other types of home visiting programs or to other participant populations need to be cautioned.

Second, the sample size, determined by available data in the intervention group of an RCT, may be smaller than optimal for latent profile analysis with a relatively large number of social relationship indicators (Tein et al., 2013). Until replicated, the identification of profiles and the profile-service involvement associations should be seen as largely exploratory. We did not apply corrections to multiple testing and did not examine if our findings differed across program sites due to the small number of participants within each site. Future studies should also examine if differences in outcomes that were skewed and with limited variability (e.g., bonding) may be detected in large sample sizes or with improved measures.

Third, we did not account for some important covariates. For example, we did not have data available to consider provider-level and program-level factors that might have influenced participant involvement, such as doulas qualifications, attachment styles and outreach strategies, or organizational-level funding, staff turnover, and levels of supervision (McFarlane et al., 2010; McGuigan & Gassner, 2016). In particular, this study did not obtain the perspectives of the doulas, which is important given that the compensation mechanism is in part based on the idea that doulas actively target and spend more effort building relationships with and providing services to mothers whom they perceived in greater need of services (e.g., Olds & Korfmacher, 1998). Insights generated from both the mother's and the home visitor's perspectives could help strengthen equitable partnerships in home visiting services. Future studies should explore if service involvement mechanisms may depend on provider- and program-level factors and participants' characteristics beyond social relationships.

Fourth, the doula services examined here were provided in the context of a novel home visiting program model in which home visits were made by multiple staff, including a doula. Although doulas had the primary contact with families during this period of study, mothers were also developing alliances with their other home visitors during pregnancy and were expecting a transition in contact from the doula to the home visitor after birth. Future research on this model should examine how working with two service providers and the shift from one service provider to another may impact changes in mothers' working alliances and service participation.

Fifth, a small number of mothers with missing data on social relationships with parent figure or father of baby were not included in this study. They may represent an extremely vulnerable group such as those without a parent figure or whose partner is deceased or incarcerated, which warrants future research. Last, qualitative methods or mixed methods can be used to explore mothers' experiences and perspectives on their relationships with service providers, their changing needs for services, their motivations, stressors, and barriers associated with the birth of their baby, and challenges to access support from families and service providers from pregnancy to the postpartum period.

6. Conclusions

Despite these limitations, this study examined different time points (e.g., pregnancy and postpartum) and different domains in service involvement (i.e., engagement and participation) and provided useful frames for home visiting service designers and providers to appreciate the important variability in the relationship history of participating mothers. This study demonstrated that different relationship backgrounds may be associated with varying proclivities to engage with services. Mothers with strong social relationships may be able to be involved in services right from the beginning of services, and mothers with challenging relationships, especially with their caregivers in childhood and current parent figures, may engage more slowly, utilizing more services over time. This study also suggested the complexity of service participation for young women without the strong support of parent figures: It is possible that they became more able to trust service providers in the process of the relationship-based interventions, but it is also possible that they became more involved in services postpartum to compensate for the lack of support from their parent figures to deal with challenges from motherhood. More research is needed to understand these dynamics.

Programs may be more effective if they are designed to be responsive to mothers' relationship histories with ongoing training and supervision. Special attention needs to be given to mothers with difficult relationships with their caregivers in childhood and current parent figures. It is not uncommon that home visitors feel frustrated with mothers who are hard to work with at first, but the data suggest that, given time, patience, and support, these mothers may open up and benefit significantly from services. For young mothers with complex histories of family relationships, home visitors can provide a safe, secure and "holding" environment to support the mothers and their young children (Ispa et al., 2006; Weatherston et al., 2020).

More importantly, the findings of this study should be understood in a broader context of systemic and structural racism and socioeconomic disadvantages that have excluded these young mothers and their families from equitable access to opportunities, disrupting positive family relationships, and sowing distrust of service systems that can present as authoritarian and unhelpful. Mothers with difficult relationships with their own families or with human services should not be pathologized. A certain reluctance to connect with service providers may reflect resilient coping with systemic and structural racism; in other words, young mothers may have learned to use distrust "as armor to preemptively protect themselves" (Levine, 2013). Future research can examine how young mothers' previous experiences of racism and discrimination may shape their service involvement processes and how home visiting programs can use culturally responsive, strength-based, and anti-racist approaches to reduce racial and ethnic health disparities (Lewy & Casau, 2021). Program designers and service providers need to go beyond the standard one-size-fits-all service protocol and value the voices and knowledge of young mothers, bring them into service design and delivery, and build equity partnerships (Singleton et al., 2022).

CRedit authorship contribution statement

Yudong Zhang: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. **Renee Edwards:** Conceptualization, Data curation, Investigation, Resources, Writing – review & editing. **Jon Korfmacher:** Investigation, Writing – review & editing. **Sydney Hans:** Conceptualization, Funding acquisition, Investigation, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This project was funded by 2019 Dolores Norton Research Award from Illinois Association for Infant Mental Health (ILAIMH) and award D89MC23146 from the MIECHV competitive grant program from the Health Resources and Services Administration (HRSA) to the State of Illinois Department of Human Services (IDHS). The contents of this publication are solely the responsibility of the authors and do not represent the official views of ILAIMH, HRSA or IDHS. The authors would like to thank their partners at the Ounce of Prevention Fund, the Illinois Governor's Office of Early Childhood Development, and the agencies that implemented the doula home visiting and case management interventions. The authors thank project director, Linda Henson, data base manager, Marianne Brennan, and the research staff involved in collecting the data, including Tanya Augustine, Melissa Beckford, Ike-sha Cain, Adriana Cintron, Nicole Dosie-Brown, Tytannie Harris, Morgan JohnsonDoyle, Katarina Klakus, Natasha Malone, Jasmine Nash, Erika Oslakovic, Jillian Otto, Amy Pinkston, Magdalena Rivota, Rosa Sida-Nanez, Luz Silva, Caroline Taromino, Ardine Tennial, Maria Torres, Yadira Vieyra, and Lauren Wilder. We are deeply appreciative of our study families for their participation.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.childyouth.2023.107228>.

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